

ANDREW M. PARKER, FORREST W. CRAWFORD, GERALD L. EPSTEIN, SELLA NEVO, DAVID GLICKSTEIN, MATTHEW L. NICOTRA, HENRY H. WILLIS

Principles for Communicating the Risks of Mirror Life

As advances in synthetic biology accelerate—enabling scientists to design biological systems from synthetic components—many researchers have begun to question whether some developments could introduce risks that outweigh their benefits. One such development involves the possible creation of *mirror life*: a conceivable alternate life form, such as a bacterium, built from proteins and nucleic acids with the opposite molecular “handedness” from those found in nature. Some scientists have warned that, if created, mirror life could pose an unprecedented global risk, because mirror bacteria could potentially evade human, animal, and plant immune systems, leading to fatal infections, and society would be unable to produce sufficient countermeasures at the needed scale. Despite the potential severity of these concerns, awareness of mirror life remains limited beyond specialist communities, and governments have yet to articulate policy responses—let alone develop strategies for communicating what mirror life is, what risks it might entail, and how close current research is to making it possible.

In this paper, we draw on research and our expertise in risk communication to outline principles that the U.S. and other governments should consider when developing strategies for communicating about mirror life to the public, research community, business community, and government agencies. We first establish a baseline



Abbreviations

DNA	deoxyribonucleic acid
DURC	dual-use research of concern
GMO	genetically modified organism
NSF	National Science Foundation
RNA	ribonucleic acid

of core facts about mirror life, including what it is, the risks it might pose if it were created, and when it might become reality. We then present the case for a government-led communications strategy and five principles for effective communication: (1) accessibility; (2) transparency; (3) use of analogies when doing so does not introduce distortions; (4) tailored messages to specific audiences; and (5) anticipation of objections, misperceptions, and inaccuracies. Finally, we outline initial steps governments could pursue to operationalize these principles.

Facts About Mirror Life, Its Development, and Potential Risks

Communicating about mirror life will require a basic understanding of mirror life. We provide an overview of what it is, the risks it could pose, when it could become reality, and what factors are driving interest in creating it. We further establish a baseline of what activities are underway within the scientific community to address this emerging risk.

What Is Mirror Life?

Most molecular building blocks necessary for life—known as *biomolecules*—have three-dimensional, asymmetric structures that are different from their mirror images, much like left hands differ from right hands. Deoxyribonucleic acid (DNA), ribonucleic acid (RNA), proteins, and many other biological molecules have this property, called *chirality*. All life on Earth is composed of biomolecules with only one of these two possible orientations. A *mirror organism* created through synthetic biology techniques would have molecular components that are mirror images of those found in nature and would constitute mirror life. For example, all known life uses proteins built from left-handed amino acids, whereas nucleic acids—DNA and RNA—are constructed from right-handed sugars. A mirror organism, by contrast, would rely on right-handed proteins and left-handed nucleic acids: a mirror image of natural biology.

Although mirror life could theoretically include multicellular organisms, such as a mirror dog or cat, scientists believe that the first instance of mirror life is likely to be a *bacterium*—a single-cell organism—because of its relative biological simplicity.

What Risks Could Mirror Life Present to Humanity and the Environment?

In a recent publication, a group of experts and concerned scientists argued that the creation of mirror life would hold significant risks for human, animal, and plant life, as well as the broader ecosystems of Earth (Adamala, Agashe, Binder, et al., 2024).

Many scientists worry that, because natural life has never encountered mirror life before, existing biologi-

Most molecular building blocks necessary for life—known as *biomolecules*—have three-dimensional, asymmetric structures that are different from their mirror images, much like left hands differ from right hands.

cal defenses would struggle to fight it effectively. This is because the opposite chirality of mirror bacteria would likely disrupt or impair the immune defenses of humans, plants, and animals. Moreover, mechanisms that normally limit the growth of bacteria in the environment—viral infection and predation—also would likely be impaired by the opposite chirality of mirror bacteria. In the absence of effective immune defenses or natural predators, mirror organisms might spread widely, causing fatal infections among many or most multicellular organisms.

Concerned scientists also question whether society could develop and deploy countermeasures sufficient to protect humanity if mirror bacteria were created. Although some existing antibiotics are not sensitive to chirality and would thus be expected to kill mirror bacteria, most existing antibiotics are designed to fit the biology of natural organisms and would probably be ineffective against mirror bacteria. It might be possible to make new antibiotics targeting mirror bacteria, but scientists doubt such countermeasures could be developed, manufactured, and distributed at the speed and scale necessary to prevent irreversible harm. Moreover, even if society were able to

protect humans with medical countermeasures, this effort would do little to mitigate potentially catastrophic impacts on plants and animals that underpin global ecosystems and food systems.

Given that mirror life does not yet exist, scientists cannot be certain about the magnitude of these risks. The only way to directly assess them would be to create mirror organisms and test their survivability in the environment and against immune systems and pharmaceuticals. This, however, presents a dangerous paradox: If mirror life is as hazardous as theorized, the very act of testing it could create the risk such experiments are intended to assess. For this reason, many concerned scientists believe that the most effective defense against mirror life is to prevent its creation; by the time its risks are empirically demonstrated, it may be too late for humanity to avoid catastrophe.

Although scientists generally view the creation of mirror life as posing potentially catastrophic risks, they tend to agree that research focused on mirror biomolecules—mirror DNA and mirror proteins—does not, on its own, carry comparable dangers. Mirror therapeutics are being pursued today because they offer some

advantages over normal (nonmirror) drugs. Efforts to explore their potential as therapeutics and diagnostics constitute legitimate research and should continue. Should decisionmakers consider how best to address the risks of mirror life, they will need to distinguish research on mirror biomolecules from efforts that assemble these components into a mirror organism.

When Could Mirror Life Become Reality?

Mirror life does not currently exist. However, research in mirror biology (and synthetic biology more generally) has made it conceivable to some scientists that a living mirror cell could be created within ten to 30 years. Some scientists estimate that, within a decade, new tools from synthetic biology could enable the construction of a living cell from synthetic nonliving parts, and following the same techniques with mirror parts could potentially produce a synthetic mirror cell (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024).

Research in mirror biology has made it conceivable to some scientists that a living mirror cell could be created within ten to 30 years.

Researchers have already synthesized mirror versions of important biomolecules, such as mirror DNA and mirror proteins. Furthermore, scientists have made partial progress toward certain steps or processes that will likely be required to create mirror bacteria, such as chemically synthesizing a genome and recoding a bacterial genome (Hutchison et al., 2016; Fredens et al., 2019). However, producing a viable, self-reproducing mirror organism likely remains years or decades away (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024).

Because mirror life would be a human-made life form, its emergence would depend on the scale and persistence of scientific effort, including funding, personnel, and institutional support. Publicly available evidence suggests that only a small number of research teams worldwide has pursued advanced work in mirror biology, and that, as of June 2026, there are no active research programs with the explicit goal of creating mirror life (Paris Conference on Risks from Mirror Life, 2025).

Advances in synthetic biology, biotechnology, or artificial intelligence could significantly accelerate progress and reduce the financial barriers to developing mirror life. One expert has estimated that a mirror cell could be developed for approximately \$500 million over the next decade, assuming a direct and transparent effort (Kubota, 2024). This cost could decrease because of improvements in biotechnology or new discoveries in synthetic biology.

Why Is There Interest in Mirror Life?

Many experts think, based on current knowledge, that the risks of mirror life outweigh the benefits. That said, some scientists have pursued advances in mirror biology to

explore potential diagnostic or therapeutic benefits. They have proposed commercial or medical uses—not for mirror life itself but for its subcellular components, such as mirror proteins or mirror DNA. These mirror biomolecules could, for example, improve diagnostics or therapeutics by reducing immune system responses and increasing stability in the body because of their mirror chirality (Harrison et al., 2023; Schearer et al., 2024).

By contrast, mirror life—a distinct and more advanced concept—refers to self-sustaining cells that assemble mirror versions of these components. Scientists have hypothesized few beneficial uses for such organisms (Adamala, Agashe, Binder, et al., 2024). Although no current projects to create mirror life are known to be underway in the United States or elsewhere in the world, the U.S. government in the past has funded fundamental synthetic biology research aimed at pursuing mirror life (National Science Foundation [NSF], 2014a; NSF, 2014b; NSF, 2019). Given the limited perceived practical benefits of synthesizing a mirror cell, actors who pursue it have been motivated by broader, fundamental goals. For example, synthesizing a mirror cell would constitute a major scientific breakthrough, and some researchers believe that the discovery may reveal insights into the chiral origins of life on Earth (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024).

How Is the Scientific Community Addressing Potential Risks?

Concerned about the potential risks of mirror life, international scientists and experts are increasingly recommending that research intended to create mirror life not be per-

mitted (Adamala, Agashe, Belkaid, et al., 2024). In January 2025, the UK Government Office for Science convened an expert roundtable to advise policymakers on the feasibility, potential benefits, and risks of mirror life, including self-replicating mirror cells, and it recommended taking steps to prevent such creation (UK Government Office for Science, 2025). At recent conferences at the Asilomar Conference Grounds in Pacific Grove, California; the University of Manchester, United Kingdom; and the Institut Pasteur in Paris, France, hundreds of experts shared the concerns that have been raised about mirror life and endorsed the need for governance (The Spirit of Asilomar and the Future of Biotechnology Summit, 2025; Institut Pasteur, 2025; University of Manchester, 2025). The National Academies of Sciences, Engineering, and Medicine (undated) held a workshop in September 2025 focused on risk assessment, governance options, and advancing responsible science domestically and internationally. In September 2025, an independent investigation from Germany’s Central Commission for Biological Safety endorsed the severity of the risk, and the UNESCO International Bioethics Committee proposed a precautionary global moratorium on creating mirror cells (Zentrale Kommission für die Biologische Sicherheit, 2025). The Carnegie Endowment for International Peace hosted a gathering of scientific, policy, and security experts, who recommended that the scientific community clearly and accurately communicate the risks of mirror bacteria to key stakeholders, that funders refrain from supporting research aimed at creating mirror life, and that policymakers engage diverse stakeholders to establish policies preventing its creation (Cuéllar, Hinderstein, and Relman, 2025). RAND publications—*Policy Options to Prevent the Creation of Mirror Organisms* (Epstein, Craw-

ford, and Nevo, 2025) and *A U.S. Strategy to Prevent the Creation of Mirror Life* (Kalra et al., forthcoming)—provide recommendations for addressing the potential threat from mirror life. Together, these initiatives aim to inform policy deliberations and promote safe research while mitigating catastrophic risks of mirror life.

Not all scientists agree on the timing of policy measures to mitigate the risks of mirror life, and some have stated that a ban on mirror life is premature because its creation remains a distant prospect (Peplow, 2025; Service, 2024). However, given the potential for catastrophic consequences—even if unlikely in the near term—early dialogue on how governments might implement policies to prevent mirror life, including how they communicate its risks to the broader public, is warranted.

Why and How Should the U.S. Government Pursue a Communications Strategy on the Risks of Mirror Life?

Should governments begin implementing policies to prevent the creation of mirror life or mitigate its risks, they will likely require a communications strategy that effectively engages the public and key stakeholders. Policy success may depend on the public's willingness to act in coordinated and specific ways, such as adopting new norms or accepting restrictions on certain activities. Building public understanding and addressing misinformation could help build trust in government officials and legitimacy for policy actions, helping with policy implementa-

tion and potentially combating politicization. Likewise, policies targeting private industry and academic institutions to prevent mirror life may require leaders to accept constraints on research and development. Such collective action requires not only clear guidance but also broad understanding and trust.

Achieving this effort will demand a communications strategy attuned to how people and organizations perceive and respond to risk, shaped by complex and interconnected analytic, emotional, and social processes. For instance, when the public perceives risks as deeply uncertain or dreaded—as in the case of mirror life, which may be seen as uncontrollable, unjust, or threatening to future generations—people may exhibit stronger, more polarized reactions, jeopardizing coordinated responses (Slovic, 2020; Slovic, 1987). To be effective, communications must therefore account for individual traits that influence how people assess risks and proposed responses, including their educational backgrounds, personalities, ethical views, community norms, lived experiences with risk, their trust in authorities, and their sense of personal and economic capacity to act (Fischhoff, 1985). Similarly, communications aimed at organizations, such as businesses or research institutions, must consider what incentives or factors influence their decisionmaking, such as pursuing prestige, generating profits, or advancing social well-being. Understanding how these factors influence risk perceptions and decisionmaking can help shape a communications strategy that supports the public's informational needs for making informed, constructive decisions (Fischhoff, 1995; Slovic, 1987; Slovic, 2020; Fischhoff, Slovic, and Lichtenstein, 1982; Wachinger et al., 2013).

Five Principles for Communicating Risks of Mirror Life

In this paper, we outline five communication principles for (1) explaining mirror life and its risks and (2) framing policy proposals to guard against those risks (see Figure 1). These principles can help guide risk communicators in speaking with the public, scientists, businesses, and government agencies that could be tasked with implementing policies to respond to the threat of mirror life. We provide recommended language for describing the science of mirror biology, describe potential analogies to other kinds of threats while carefully noting when these analogies may be counterproductive, and offer ways of addressing confusion and concerns about mirror life, research into mirror life, and governance proposals.

Our recommendations are based on the technical findings about the risks of mirror life (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024), as well as research on effective risk communication about threats to public health and security (Bartolucci et al., 2023; Fischhoff, 1995; Fischhoff, Brewer, and Downs, 2011; Pidgeon, 2021; Steratore et al., 2023; Warren and Duckett, 2026).

Technical aspects of mirror life development, detailed timelines to mirror life creation, and specific policy recommendations are beyond the scope of this paper.

1. Be Clear by Describing the Facts and Risks in an Accessible Way

A core principle of risk communication is bridging the gap between what people currently understand and what they need to know to make informed decisions about risk (Morgan et al., 2001). Early statements on mirror life should quickly orient audiences to the key facts, terminology, and potential risks of mirror life. When communicating the scientific facts about mirror life to nonspecialists, messengers should use plain language and avoid jargon and complex scientific descriptions (Fischhoff, Brewer, and Downs, 2011).

Messengers should opt for simpler language and avoid inaccessible terms that are likely unfamiliar to most people. For instance, instead of saying “enantiomer,” communicators can use “mirror-image molecule.” Table 1 provides several accessible vocabulary alternatives relevant to mirror biology that may aid public discussion. While these terms

Using plain language to describe mirror life

Most molecules needed for life, such as proteins and DNA, have three-dimensional structures that differ from their mirror image. Life on Earth uses only one orientation of these molecules. In the future, scientists might be able to create a mirror cell in a lab. It would work like a natural cell but would have proteins and DNA shaped in the opposite way.

FIGURE 1

Communication Principles for Explaining Mirror Life to the Public

1

Be clear

Describe the facts and risks in an accessible way, without jargon.

2

Be transparent

Be open about what is known and unknown about the risks and where value judgments must be made, refrain from masking uncertainty, and emphasize the importance of alternative viewpoints.

3

Use analogies

Guide audiences toward more-useful analogies; in circumstances in which analogies are not counterproductive, make sure to explain their limitations.

4

Tailor to audience

Craft strategies to provide understandable and actionable guidance relevant for different audiences with distinct priorities, knowledge, and context, and then test communication needs and messages among the intended audiences.

5

Anticipate objections, misperceptions, and inaccuracies

Preemptively acknowledge commonly challenged or misunderstood views and actively counter false information by providing explanations of inaccuracies and alternative viewpoints.

TABLE 1
Potential Accessible Alternatives to Terms
That May Emerge in Technical Discussions of
Mirror Bacteria

Inaccessible	Examples of Potential Accessible Alternatives
Chirality	Handedness; a feature of an object that makes its mirror image different from the original
Chiral molecule	A molecule with an asymmetric, three-dimensional shape, like left and right hands; a left- or right-handed molecule
Amino acid, peptide	Building block or piece that makes up proteins; a protein unit
Antigen	A substance that an immune system reacts to; an immune system trigger
Antibody	A molecule that helps the immune system fight foreign substances; an immune system defender
Auxotroph	An organism designed to be dependent on nutrients that are only provided in the lab
Immunogenic	Capable of causing an immune system response; immune system-activating
Enantiomer	One of the two possible mirror-image forms of a molecule
Genome	An organism's genetic material; the full set of DNA or RNA
Nucleic acid	DNA or RNA
Nucleotide, oligonucleotide	Building block or piece that makes up DNA or RNA; a DNA or RNA unit
Ribosome, polymerase, etc.	Molecules in cells that perform specific roles; molecular machines
Phage, bacteriophage	Virus that infects bacteria

offer useful starting points for initial conversations, further research should assess their accessibility and identify additional definitions or descriptions best suited for informing the public.

2. Be Transparent About Risks, Value Judgments, and Uncertainty

Communicators of the risks of mirror life should strive for transparency with the general public and other audiences, because transparency increases the credibility of communication and conveys respect for the audience (U.S. Environmental Protection Agency, 2025; Federal Emergency Management Agency, 2024; Fischhoff, Brewer, and Downs, 2011). They should be open about the current status and trajectory of mirror life development, the uncertainty of its risks and policy responses, any value judgments, the extent of scientific consensus, and key context surrounding the underlying evidence.

Status and trajectory: Messengers should aim to clearly share what is known regarding progress in mirror life research and what developments might unfold in the future (Fischhoff, 1995; Fischhoff, Brewer, and Downs, 2011). In current communications, they should stress that mirror life does not yet exist but could be created in the next ten to 30 years. Communicators should be aware that different people may interpret verbal probabilistic statements (e.g., the term *unlikely*) differently. Best practices suggest incorporating quantitative estimates or ranges where possible (Budescu and Wallsten, 1995; Wallsten et al., 1993). That said, many current uncertainties may be profound, preventing numeric estimates.

Being clear about the current status and trajectory of mirror life development

Mirror life does not currently exist. However, research programs have made progress in creating some of the mirror molecules that could one day be used to create a mirror cell. Scientists are not likely to create mirror life in the next ten years. Beyond that, the timeline is unknown.

Uncertainty of risks and policy responses: Communicators should be open with the public about uncertainties surrounding the risks of mirror life and the potential effectiveness of policy responses, avoiding exaggeration. They can frame these uncertainties as an opportunity to reinforce their commitment to discourse and openness to new evidence (Fischhoff, 1995).

Value judgments: Communicators should, in general, aim to avoid imposing value judgments in their messaging to the public. An important goal of risk communication is to inform and empower people to make decisions based on their own values and support policy discussions that reflect those values. This goal contrasts with communication intended to influence or pressure people to accept a decision or policy (Fischhoff, 1995). However, the individuals communicating risks are often the very same people, such as political or government leaders, who are responsible for implementing policies that rest on value judgments, such as the need to save as many lives as possible. In these cases, communicators should, at a minimum, be transparent about values underpinning their messaging and policy proposals.

Extent of scientific consensus: To further ensure transparency, communicators should clearly convey the extent of scientific consensus and attribute claims directly to sources. For example, highlighting broad agreement among scientists about the risks of mirror life can help convey the credibility of these risks. Conversely, when consensus is lacking, acknowledging differing perspectives can foster public trust and reinforce the legitimacy of the messaging (Fischhoff, 1995; Fischhoff, Brewer, and Downs, 2011).

Context surrounding evidence: When relevant, communicators should be transparent about the U.S. government's relationship to specific claims and processes. For example, communications might explain that the U.S. government learned about possible mirror life threats from external scientists, that those scientists made the initial announcement, and that the government is now evaluating those claims. Messengers should clarify the level of credibility assigned to such claims, whether by the U.S. government as a whole or by specific agencies.

Communicating the extent of scientific consensus

A group of experts and concerned scientists warns that creating mirror life could be dangerous for humans, animals, and plants. Most scientists who have worked in mirror life research recommend halting efforts to create it.

Communicating uncertainty and credibility of evidence

Mirror bacteria could spread widely. Their mirror image might protect them against immune systems, medicines, or natural enemies.

Because mirror life does not yet exist, the risks are not certain. However, many experts agree that they are serious enough to take precautions.

3. Guide Audiences Toward Analogies When Useful and Explain Their Limitations

To help the public understand mirror life, its risks, and potential policies, messengers can use analogies to more familiar or similar-seeming technologies and risks. When first introduced to the topic, audiences are likely to anchor their opinions to these comparisons (Bostrom, 2011; Gentner and Holyoak, 1997). The news media, which often relies on analogies as powerful explanatory tools, can help reinforce these initial impressions (Edwards, 2003), as can social media.

However, using analogies also presents pitfalls. The threats potentially posed by mirror life are unique and unprecedented, so there are no close analogies. Any analogy

will inevitably have aspects that do not apply to mirror life, meaning that drawing on those analogies risks distorting understanding, whether intentionally or not (Bostrom, 2011). Therefore, communicators should pair analogies with clear explanations of how mirror life differs from the example.

We recommend guiding audiences toward more-illuminating and more-useful analogies, avoiding misleading or less relevant ones, and being prepared to discuss the limitations of analogies that arise (Bruine de Bruin, Carman, and Parker, 2021). Useful analogies do not need to be biological or health related, for example, if they provide helpful policy solution parallels. No analogy will be perfect; some may be accurate in their characterization of a threat but inaccurate in their implied set of solutions.

Communicators should anticipate audiences forming their own analogies. In such cases, they should avoid ignoring or summarily dismissing them, which may backfire, and instead use this as an opportunity to both head off analogies commonly referenced by audiences that do not align well with actual risk while promoting better-aligned analogies (Bruine de Bruin, Carman, and Parker, 2021).

Analogizing the Risks of Mirror Life

Analogies can help the public grasp the risks and scale of mirror life (Bostrom, 2011; Gentner and Holyoak, 1997). For instance, communicators might compare its potential societal impacts with those of drug-resistant superbugs (see Figure 2), invasive species (see Figure 3), or even non-biological catastrophic risks, such as nuclear weapons or climate change.

By comparing mirror life with superbugs, communicators can help the public understand how mirror life could spread rapidly, be difficult or impossible to treat, and require global cooperation to manage its risks. However, because the analogy is imperfect, it may risk leading the public to incorrect conclusions (for instance, that mirror life could originate spontaneously from nature). Therefore, communicators should clarify these distinctions, emphasizing that mirror life requires deliberate human engineering and that many experts support preventing its creation.

Other possible biology-related analogies include the following:

- **Genetically modified organisms (GMOs):**

Although it is almost certainly a poor analogy, some members of the public might compare mirror life to GMOs, given that both result from human interven-

tion and do not naturally occur in the environment (U.S. Food and Drug Administration, undated).

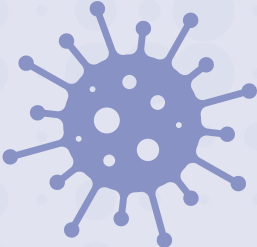
Communications surrounding this analogy should emphasize that public perceptions of GMO-related risks often differ significantly from expert assessments and, in any case, do not convey mirror life's potentially catastrophic risks to society.

- **Prions:** Both prions and mirror life involve unusual molecular structures that defy traditional biology, are difficult to detect, and pose serious, poorly understood health risks with no known cure. *Prions*—responsible for such diseases as bovine spongiform encephalopathy (BSE) and variant Creutzfeldt-Jakob disease (vCJD)—are misfolded proteins that induce similar misfolding in other proteins (U.S. Centers for Disease Control and Prevention, undated). However, prions are likely far less transmissible than mirror life could be, cause harm through a very different mechanism, and remain rare in humans.

Analogies might also draw from other, nonbiological domains that are likely to be broadly familiar to the public, such as the following:

- **Ozone depletion and climate change:** These environmental challenges offer a useful analogy for mirror life in terms of risk management: Both involve human-caused threats that likely require sustained global cooperation to address. However, these analogies are less effective when considering how the risks originate. Ozone depletion and climate change stem from countless individual and institutional actions around the world, while mirror life could emerge

FIGURE 2
Mirror Life vs. Drug-Resistant Superbugs

Drug-Resistant Superbugs	How They Are Similar to Mirror Life	
	• Both are biology-based threats that could spread rapidly and be difficult or even impossible to treat. • Both could be theoretically created in a single laboratory. • Both could demand sustained global cooperation for prevention and mitigation.	
	How They Are Different from Mirror Life	
 Superbugs are bacteria, viruses, or fungi that are resistant to the most common antimicrobial medicines used to fight infections. Examples include methicillin-resistant <i>Staphylococcus aureus</i> (MRSA), vancomycin-resistant <i>Staphylococcus aureus</i> (VISA), and extensively drug-resistant tuberculosis (XDR-TB).	Superbugs	Mirror Life
	• Already exist and can arise naturally; emerged from routine use of medicines, such as antibiotics • Are a widespread danger, primarily to humans • Do face some existing countermeasures • Are often studied to help treat superbug infections	• Does not yet exist; it will not arise without human engineering through concerted scientific effort • Could potentially cause the complete devastation of most life and be far worse than superbugs • Has no confirmed countermeasures; superbug countermeasures are unlikely to work against mirror life • Experts recommend against creating mirror life, including treatment research

SOURCE: Features information from National Institutes of Health, 2014.

Invasive Species	How They Are Similar to Mirror Life					
 Invasive species are non-native organisms (usually plants and animals) introduced to an ecosystem that can cause harm to native organisms or the local environment.	- Both have the potential to spread rapidly by evading predation by native organisms. - Both could affect humans, animals, and plants and could contaminate soil and water sources. - Both could kill native organisms by outcompeting them for resources.					
	How They Are Different from Mirror Life					
	<table> <tr> <th data-bbox="628 688 1062 757">Invasive Species</th><th data-bbox="1062 688 1418 757">Mirror Life</th></tr> <tr> <td data-bbox="628 757 1062 920"> - Can disrupt ecosystems and cause death or extinction of some species - May be vulnerable to some predation and defenses from local species - Are limited in the extent of their propagation </td><td data-bbox="1062 757 1418 920"> - Has the potential for complete devastation of life; could be far worse than invasive species - Is unlikely to be vulnerable to the vast majority of predation and defenses; potentially unrecognizable or undetectable to natural immune systems - Could spread widely with no known limits </td></tr> </table>	Invasive Species	Mirror Life	- Can disrupt ecosystems and cause death or extinction of some species - May be vulnerable to some predation and defenses from local species - Are limited in the extent of their propagation	- Has the potential for complete devastation of life; could be far worse than invasive species - Is unlikely to be vulnerable to the vast majority of predation and defenses; potentially unrecognizable or undetectable to natural immune systems - Could spread widely with no known limits	
Invasive Species	Mirror Life					
- Can disrupt ecosystems and cause death or extinction of some species - May be vulnerable to some predation and defenses from local species - Are limited in the extent of their propagation	- Has the potential for complete devastation of life; could be far worse than invasive species - Is unlikely to be vulnerable to the vast majority of predation and defenses; potentially unrecognizable or undetectable to natural immune systems - Could spread widely with no known limits					

from a single laboratory. Additionally, environmental risks are generally more gradual in impact, less conceivable to result in human extinction, and more reversible than the potential consequences of mirror life (Adamala, Agashe, Belkaid, et al., 2024; UK Department for Environment, Food & Rural Affairs, undated).¹

- **Nuclear weapons:** Both nuclear weapons and mirror life carry the potential to cause massive and catastrophic destruction, making nuclear weapons a potentially effective analogy for illustrating mirror life's possible devastation.² However, unleashing nuclear devastation at a global scale likely requires exquisite intentional use of a nuclear weapons arsenal sized on the order of the peak of global nuclear weapon inventories in the 20th century,³ whereas triggering the effects of mirror life might, in the long run, only require the actions of one or a small group of actors. It is also technically possible to have a geographically limited use of nuclear weapons—one that would not be globally catastrophic. Spread of mirror organisms may not be geographically bounded. Furthermore, unlike nuclear weapons, mirror organisms have not been created and tested, so there is greater uncertainty about the impacts that the release of mirror bacteria would cause.

Communicators can also draw on fictional or hypothetical scenarios. Often not grounded in real-world threats, such fantastical narratives may nevertheless have shaped perceptions among the audience about the nature of global risk, and they may help convey the potential scale of devastation posed by mirror life, particularly given the absence of direct historical parallels. However, communicators should

be cautious not to overstate these fictional cases' similarities to mirror life, articulating differences to maintain trust and credibility. Possible examples include the following:

- **"Gray goo":** This theoretical scenario, in which self-replicating nanorobots escape human control and turn all matter into copies of themselves, can serve as an analogy for mirror life, illustrating how uncontrolled replication of a novel life form could pose existential threats (Drexler, 1992).
- ***The Andromeda Strain*:** This novel by Michael Crichton (1969) describes a microscopic alien life form that can propagate and cause harm—not by specifically attacking normal living systems in the way natural pathogens would, but by reproducing in ways that harm those normal living systems. In the story, the strain behaves similarly to mirror life: Human immune systems cannot fight it, and ordinary medical countermeasures are ineffective.
- **"Ice-nine":** From Kurt Vonnegut's *Cat's Cradle* (1963), this fictional substance crystalizes liquid water on contact. In the story, scientists realize the use of the substance would solidify the oceans and kill all life on Earth. Fictional ice-nine is not a living organism, but its rapid spread has catastrophic consequences from a single point of contact with the ocean.
- ***Jurassic Park*:** This 1990 Crichton novel, and subsequent film, depicts scientists reviving dinosaurs using DNA recovered from fossilized amber, supplemented with modern frog DNA to fill genetic gaps. Although the project's founder pursues de-extinction out of scientific curiosity, prestige, and profit, the breakthrough leads to unintended, cata-

strophic consequences when the dinosaurs unexpectedly reproduce. This dynamic reflects concerns about mirror life: Once new life can replicate, containing its spread might prove impossible.

Table 2 serves as a reference tool to help communicators quickly assess which aspects of commonly used analogies are similar to mirror life and where key differences lie. By outlining points of overlap and distinction, this table

TABLE 2
How Mirror Life Compares with Other Risks

Category	Characteristic	Mirror Life	Drug-Resistant Superbugs	Invasive Species	GMOs	Prions	Depleted Ozone/Climate Change	Nuclear Weapons
Status	Theoretically possible	✓	✓	✓	✓	✓	✓	✓
	Currently nonexistent	✓						
Origin	Stems from biology	✓	✓	✓	✓	✓		
	Manmade	✓			✓		✓	✓
	Can be created in a single lab	✓	✓		✓	✓		
Effects	Unknown risks	✓						
	Effects could be irreversible	✓	✓	✓	✓	✓	✓	✓
	Potential for rapid spread	✓	✓	✓	✓			
	Can also affect plants and animals	✓	✓	✓	✓		✓	✓
	Can cause total global devastation	✓					✓	✓
Governance and use	Requires ongoing global cooperation	✓	✓	✓	✓	✓	✓	✓

can guide more accurate risk communication and analogy selection.

Analogizing Policy Responses to Mirror Life

In addition to illustrating potential risks, analogies can help explain policy options for preventing mirror life–related threats to civilization. For instance, in describing a pause or prohibition on research that would enable or

create mirror life, communicators may reference efforts to manage other catastrophic risks, such as a ban on weather weapons during the 1970s (see Figure 4). This example highlights a past international initiative in which states agreed to prohibit technologies before they exist, as would be the case with a mirror life ban. However, communicators should articulate a clear difference: Whereas motivations to produce weather weapons would likely have been

FIGURE 4
Weather Weapon Policy as an Analogue for Managing Mirror Life

Weather Weapons	
	How It Could Apply to Mirror Life
	Policies addressing the threat of mirror life could involve international agreements, such as those used to ban weather weapons. The Environmental Modification Treaty bans the “military or any other hostile use of environmental modification techniques having widespread, long-lasting or severe effects as the means of destruction, damage or injury to any other State Party.” Similar to mirror life, weather weapons did not exist at the time of the ban (and still do not), but if they did, they would likely pose “widespread, long-lasting or severe” risks.
	Why It Might Not Apply to Mirror Life
	The destructive potential of mirror life could far exceed any particular use of a weather weapon, if such existed. That said, the primary concern today centers on the development of mirror life for scientific purposes rather than as a weapon. International agreement aimed at prevention would likely apply to all research that would enable or create mirror life and not be limited to military applications.

SOURCE: Features information from the Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques, 1977.

driven by military objectives, creation of mirror life may result from civil or scientific goals, and coordinating a global response to a perceived civil matter might prove more difficult.

Rather than an outright global ban on mirror life research, policymakers could pursue a narrower approach, such as controls on dual-use research of concern (DURC) (see Figure 5). When drawing this analogy, communicators

FIGURE 5
DURC Policy as an Analogue for Managing Mirror Life

Dual-Use Research of Concern (DURC)	How It Could Apply to Mirror Life
 Key U.S. policies aim to govern DURC, which encompasses scientific research with both significant societal benefits and the potential for serious misuse. Per these policies, the government conducts case-by-case review of proposals to weigh the potential risks and benefits and decides whether it is worthwhile to conduct or fund the research and how to mitigate risks.	As the government's mechanism for overseeing potentially risky research, DURC policies could naturally extend to research that could enable the creation of mirror life.
	Why It Might Not Apply to Mirror Life Unlike much dual-use research that promises significant societal benefits—such as advances in prescription medications—the development of mirror life presents risks that far outweigh any potential gain. For that reason, policy proposals to address research that would create mirror organisms would likely take the form of comprehensive research or funding bans. However, much of the research that could enable the eventual creation of mirror organisms—even if that were not its objective—may have value in other applications. Governing this type of research will likely require the establishment of red lines beyond which research would not be permitted, together with the need for case-by-case review and approval of research that approaches or circumvents those lines.

SOURCE: Features information from Administration for Strategic Preparedness and Response, undated.

should emphasize a key distinction: Many defend DURC as research on technologies whose benefits outweigh their risks, whereas scientific consensus suggests that mirror life research carries costs that far surpass its expected benefits.

4. Tailor Communication Strategies for Different Audiences

The creation of mirror life may require messengers to communicate about the facts, risks, and policy proposals to multiple audiences, tailoring their messaging for each. Fundamentally, risk communication surrounding mirror life will need to bridge technical and nontechnical audiences (Steratore et al., 2023). While some messages may be primarily relevant to only select groups (e.g., business, research institutions, or government agencies), other messages, such as the government's focus on public and environmental safety, will be important for everyone.

Customizing for each audience: Effective communication should address the unique informational needs of each audience (Bartolucci et al., 2023). For example, communications to the general public will likely require clear, accessible explanations of what mirror life is, why it matters, and what society can do to protect itself against risks. Within this audience, specific groups may have particular concerns, hold common misconceptions, or face knowledge gaps tied to their individual or community context (Morgan et al., 2001). For instance, environmental conservationists may worry about the potential for ecosystem devastation resulting from the accidental release of mirror life, while biotechnology investors may be concerned about

restrictive policies affecting innovation progress. Effective strategies for meeting audience needs often involve engaging with them directly (Kousky and Shabman, 2015; Pidgeon, 2021; Steratore et al., 2023; Warren and Duckett, 2026), learning their communication and decisionmaking priorities (Árvai, 2014), and codeveloping message strategy and content (Warren and Duckett, 2026). Finally, rigorous message testing with the intended audience rather than with experts, as well as ongoing monitoring and evaluation, can help align messaging with audience needs and communication preferences (Fischhoff, Brewer, and Downs, 2011; Helmus, Hiatt, and Becerra-Ornelas, 2022; Morgan et al., 2001; Steratore et al., 2023).

Deploying different messengers: Audiences respond better to credible and trusted messengers (Balog-Way, McComas, and Besley, 2020; Bartolucci et al., 2023), especially with novel threats (National Academies of Sciences, Engineering, and Medicine, 2016; Siegrist, 2021), and different audiences may respond better to different messengers. For example, scientific and business audiences may be more receptive to communications from scientists and business leaders. The general public may place greater trust in local community and political leaders, making education of these leaders a potentially valuable communication channel. In the current climate of partisan rhetoric, using a bipartisan or nonpartisan voice, if possible, may be important.

In Figure 6, we provide recommended language for setting expectations about the emergence and risks of mirror life, along with example messengers likely to resonate with each audience.

FIGURE 6

Recommended Language and Example Messengers to Set Expectations

Communicating
to the
general public



“

Today, people do not need to act on their own, because mirror life will not appear soon. It cannot form naturally, and even expert scientists cannot make it yet. However, the risks, if not refuted, are serious. Therefore, the U.S. and other governments will need new policies to manage those risks. We encourage people to stay informed on the latest developments for handling this unique challenge. —**U.S. federal public health messenger**

”

Communicating
to **federal**
agencies



“

U.S. government agencies might be tasked with implementing aspects of a public policy or regulatory response to prevent development of mirror life. Relevant agencies could include scientific funding and research agencies, regulatory agencies, and advisory bodies. The administration could also explore collaboration with foreign governments and international organizations. —**White House messenger**

”

Communicating
to the **business**
community



“

Research on mirror biomolecules might lead to useful commercial products. However, this work does not require creating mirror life—and it should not lead there. Creating mirror life for any purpose could make it much easier for someone to build a version that could end most life on Earth. Because of this risk, we are working with experts and scientists to determine how far mirror biomolecule research can go before it gets too close to creating mirror life. Many experts believe that the possible benefits of mirror life could be achieved in other, safer ways. Policy responses should stay narrow and targeted. This approach would reduce risks without slowing innovation or fair competition. Businesses should also take part in policy discussions and share what they know about the risks and the pace of progress in synthetic biology. —**Business community messenger**

”

5. Anticipate Objections, Misperceptions, and Inaccuracies

As seen in past health crises, efforts to manage risks may prompt public objections, including those based on misconceptions or misunderstandings (American Psychological Association, 2024; Bateman and Jackson, 2024). Messengers will likely need to anticipate these objections and pursue strategies that acknowledge concerns without alienating the public. Confrontational approaches, such as debating or criticizing, can backfire, prompting defensiveness or entrenched beliefs. Instead, messengers should convey respect and empathy and aim to appeal to deeper emotions that may be driving objections (Bateman and Jackson, 2024).

Additional strategies may include using clear labels to help the public identify inaccurate information, relying on appropriate and trusted messengers (e.g., experts or other nonpartisan figures) and “prebunking” by providing accurate information before objections can emerge and gain traction (Bateman and Jackson, 2024; American Psychological Association, 2024).

Although popular objections have yet to emerge in broad public discussions, we can begin to anticipate potential narratives that doubt key facts about mirror life, its risks, and possible policy responses. For instance, we anticipate some may dismiss the threat as science fiction or impossible. In response, messengers should acknowledge

that mirror life does not yet exist while highlighting expert judgments that its creation could become possible in the coming years or decades. Other objections may focus on advancing scientific achievement, delaying action until mirror life emerges, stifling legitimate research, exaggerating risks, or arguments that mirror life must be created to study countermeasures. Figure 7 provides examples of these possible objections and helpful responses.

In addition to responding to good-faith counterarguments regarding mirror life, it is important to address false or inaccurate information, especially claims intended to deceive the public. Such narratives may come from individuals who oppose policy proposals to address the threat from mirror life or from adversaries seeking to exploit uncertainty around the issue. Figure 8 summarizes possible narratives and helpful responses.

Finally, anticipating and addressing objections, misperceptions, and inaccuracies in an effective and timely manner requires the capacity to quickly conduct empirical research that examines the current information environment, increases understanding of how the public (or other communication recipients) interprets new information, and tests alternative communication approaches. Such research is best accomplished with pre-positioned research infrastructure (Parker, 2022; Parker et al., 2020) designed to anticipate and facilitate rapid deployment of targeted studies, rather than requiring entirely new research efforts each time.

FIGURE 7

Possible Objections to Mirror Life Information and Example Responses

Disbelief

Objection:

Mirror life is science fiction—it doesn't exist.

Response:

Scientists generally agree that mirror life could be created within the next few years to decades. There are reasons to think that its creation could be catastrophic for most life on Earth.

Scientific achievement

Objection:

Mirror life sounds cool—we shouldn't limit research on it.

Response:

We cannot create the risk just to study mirror life or prove that we can achieve it. It would be extremely difficult or even impossible to study mirror life safely and securely.

Researching mirror life could be catastrophic to society. Once created, we might be unable to control or reverse its dangers.

Delayed response

Objection:

The risks should be addressed later, when a mirror cell is closer to being realized.

Response:

By waiting until mirror cells are created, we would lose the most effective defense: not creating them at all. We should keep certain technical hurdles unsolved.

Figure 7—Continued

**Stifling
legitimate
research**

Objection:

We already have regulations that govern dangerous or dual-use research. More governance will harm commercial interests and/or legitimate and beneficial scientific research.

Response:

Oversight of dual-use research is an important—and different—problem.

Dual-use research, while often legitimate, may pose risks. Current policies, such as those for dual-use research of concern (DURC), use case-by-case review to weigh benefits against harms.

However, mirror life creation presents unique and potentially unprecedented risks. If expert assessments are correct, the risks of mirror life could far outweigh any known benefits. Existing dual-use policies are not suited for risks at this scale.

However, research into mirror biomolecules could have legitimate uses unrelated to mirror life. There would likely need to be red lines beyond which such research could not cross, maintaining barriers to the creation of mirror life.

Figure 7—Continued

Exaggerated risk

Objection:

Risks from synthesized mirror bacteria would be minimal, as they cannot consume the non-mirror nutrients found outside the lab.

Response:

A functional mirror cell could be engineered to consume abundant non-mirror nutrients, such as ordinary glucose.

Some nutrients in the environment have the same shape as their mirror image, so mirror bacteria would be able to eat them.

It is indeed likely that mirror bacteria could be created that would not pose an immediate risk. However, a bad actor could work around or remove safety measures. This can pose a catastrophic and irreversible risk.

Researching countermeasures

Objection:

We need mirror biology to create countermeasures in case others create mirror life.

Response:

There are research pathways available to create countermeasures that do not require the creation of mirror life.

By creating mirror life to develop countermeasures, we would create the very threat that we would be trying to prevent. Furthermore, many aspects of the threat posed by mirror life would not be addressed.

FIGURE 8

Possible Inaccurate Narratives About Mirror Life and Example Responses

Weaponization of mirror life

Objection:

The United States sees military applications of mirror life and wants to weaponize it before its adversaries do.

Response:

The United States is party to the Biological Weapons Convention and has categorically renounced the development or use of biological weapons.

Furthermore, mirror life would be a self-defeating weapon. It could not be controlled, and any attempt to use it would almost certainly destroy the attacker as well.

Commercialization and strategic advantage of creating mirror life

Objection:

The United States sees commercial utility in mirror life and an advantage in creating it before its competitors do.

Response:

Mirror life has been identified as an extreme threat to all complex life on Earth.

Although there might be beneficial or commercial uses for mirror biomolecules in the future, the risks of assembling an entire mirror organism are believed by experts to outweigh any potential benefits.

Given the Current Situation, What Else Can Be Done?

The public conversation about the risks of mirror life will likely continue as synthetic biology advances, increasing both technical understanding and progress toward creating mirror life. At the same time, research on the possible risks and proposals for governance and mitigation will evolve. Communication guidelines presented in this paper can help policymakers understand how risk perceptions may develop and inform the demands of risk communication.

Evolving scientific, political, and geopolitical dynamics will shape public perceptions of mirror life risks and, in turn, influence risk communication strategies. Breakthrough scientific discoveries may generate public excitement over potential benefits, while laboratory accidents, disease outbreaks, or other hazards could heighten concerns or fears. Perceived competition from adversaries seeking to be first in creating mirror life could further sway public support for mitigation policies. Communicators must monitor these shifting dynamics and adapt messaging to align with changes in public awareness and sentiment regarding risk management.

Building on the lessons in this paper, we recommend that policymakers take the following steps now to ensure that a communications strategy draws from reliable, complete, and trusted information.

1. **Establish a baseline understanding of the public's perceptions of mirror life and attitudes toward mitigation measures.** Such research will help policymakers understand how views are changing and support testing of risk communication messages.

2. **Identify the types of information policymakers and the public need to understand mirror life and weigh the advantages and disadvantages of mitigation measures.** This advance work can help messengers prepare a risk communication strategy and materials. Messengers should aim to understand common misconceptions about mirror life and how they influence risk perceptions and support for mitigations.
3. **Identify potential risk communicators and provide them with the tools for effective risk communication.** This approach includes understanding how stakeholders make sense of new risks and how they use this understanding to make decisions. It also includes strategies for engaging target audiences in two-way communication rather than simply falling back on one-way educational efforts.
4. **Prepare contingencies that might affect views on mirror organisms by using this information to develop and test communications.** Testing should involve potential audiences to ensure that such messaging is comprehensible and meets their information needs.

Research on risk perceptions and communication highlights policymakers' crucial role in informing the public about potential risks and fostering informed discussions on risk management. While the risks from mirror life may seem unimaginable to many today, following these foundational steps can help policymakers clearly explain such risks, increasing the likelihood of effective risk mitigation.

Notes

¹ According to unpublished 2024 research conducted by RAND researchers Henry H. Willis, Anu Narayanan, Benjamin Boudreaux, Bianca Espinosa, Edward Geist, Daniel M. Gerstein, Dahlia Anne Goldfeld, Nidhi Kalra, Tom LaTourrette, Emily Lathrop, Alvin Moon, Jan Osburg, Benjamin Lee Preston, Kristin Van Abel, Emmi Yonekura, Robert J. Lempert, Sunny D. Bhatt, Chandra Garber, and Emily Lawson.

² According to unpublished 2024 research conducted by Willis and colleagues.

³ According to unpublished 2024 research conducted by Willis and colleagues.

References

- Adamala, Katarzyna P., Deepa Agashe, Yasmine Belkaid, Daniela Matias de C. Bittencourt, Yizhi Cai, Matthew W. Chang, Irene A. Chen, George M. Church, Vaughn S. Cooper, Mark M. Davis, et al., “Confronting Risks of Mirror Life,” *Science*, Vol. 386, No. 6728, 2024.
- Adamala, Katarzyna P., Deepa Agashe, Damon J. Binder, Yizhi Cai, Vaughn S. Cooper, Ryan K. Duncombe, Kevin M. Esvelt, John I. Glass, Timothy W. Hand, Thomas V. Inglesby, et al., *Technical Report on Mirror Bacteria: Feasibility and Risks*, Stanford Digital Repository, December 2024.
- Administration for Strategic Preparedness and Response, “Dual Use Research of Concern Oversight Policy Framework,” webpage, undated. As of September 30, 2025: <https://aspr.hhs.gov/S3/Pages/Dual-Use-Research-of-Concern-Oversight-Policy-Framework.aspx>
- American Psychological Association, “8 Recommendations for Countering Misinformation,” last updated March 1, 2024. As of April 24, 2026: <https://www.apa.org/topics/journalism-facts/misinformation-recommendations>
- Árvai, Joseph, “The End of Risk Communication as We Know It,” *Journal of Risk Research*, Vol. 17, No. 10, 2014.
- Balog-Way, Dominic, Katherine McComas, and John Besley, “The Evolving Field of Risk Communication,” *Risk Analysis*, Vol. 40, No. S1, November 2020.
- Bartolucci, A., M. C. Aquilino, L. Bril, J. Duncan, and T. van Steen, “Effectiveness of Audience Segmentation in Instructional Risk Communication: A Systematic Literature Review,” *International Journal of Disaster Risk Reduction*, Vol. 95, September 2023.
- Bateman, Jon, and Dean Jackson, *Countering Disinformation Effectively: An Evidence-Based Policy Guide*, Carnegie Endowment for International Peace, 2024.
- Bostrom, Anne, “Lead Is Like Mercury: Risk Comparisons, Analogies and Mental Models,” in Asa Boholm, ed., *New Perspectives on Risk Communication: Uncertainty in a Complex Society*, Routledge, 2011.
- Bruine de Bruin, Wändi, Katherine G. Carman, and Andrew M. Parker, “Mental Associations with COVID-19 and How They Relate with Self-Reported Protective Behaviors: A National Survey in the United States,” *Social Science & Medicine*, Vol. 275, April 2021.

- Budescu, David V., and Thomas S. Wallsten, "Processing Linguistic Probabilities: General Principles and Empirical Evidence," in Jerome Busemeyer, Reid Hastie, and Douglas L. Medin, eds., *Psychology of Learning and Motivation*, Vol. 32, Academic Press, 1995.
- Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques, signed at Geneva, May 18, 1977.
- Crichton, Michael, *The Andromeda Strain*, Alfred A. Knopf, 1969.
- Crichton, Michael, *Jurassic Park*, Alfred A. Knopf, 1990.
- Cuéllar, Mariano-Florentino (Tino), Corey Hinderstein, and David Relman, "Prevention Beats Nonproliferation: Addressing the Risks of Mirror Life," Carnegie Endowment for International Peace, October 3, 2025.
- Drexler, K. Eric, *Nanosystems: Molecular Machinery, Manufacturing, and Computation*, Wiley, 1992.
- Edwards, Adrian, "Communicating Risks Through Analogies," *BMJ*, Vol. 327, September 2003.
- Epstein, Gerald L., Forrest W. Crawford, and Sella Nevo, *Policy Options to Prevent the Creation of Mirror Organisms*, RAND Corporation, PE-A3436-1, March 2025. As of October 1, 2025: <https://www.rand.org/pubs/perspectives/PEA3436-1.html>
- Federal Emergency Management Agency, *2024 FEMA Risk Communications, Crisis Communications + Community Engagement Summit Report*, U.S. Department of Homeland Security, August 2024.
- Fischhoff, Baruch, "Managing Risk Perceptions," *Issues in Science and Technology*, Vol. 2, No. 1, 1985.
- Fischhoff, Baruch, "Risk Perception and Communication Unplugged: Twenty Years of Process," *Risk Analysis*, Vol. 15, No. 2, 1995.
- Fischhoff, Baruch, Paul Slovic, and Sarah Lichtenstein, "Lay Foibles and Expert Fables in Judgments About Risk," *American Statistician*, Vol. 36, No. 3b, 1982.
- Fischhoff, Baruch, Noel T. Brewer, and Julie S. Downs, eds., *Communicating Risks and Benefits: An Evidence-Based User's Guide*, U.S. Food and Drug Administration, August 2011.
- Fredens, Julius, Kaihang Wang, Daniel de la Torre, Louise F. H. Funke, Wesley E. Robertson, Yonka Christova, Tiongsun Chia, Wolfgang H. Schmied, Daniel L. Dunkelmann, Václav Beránek, et al., "Total Synthesis of *Escherichia coli* with a Recoded Genome," *Nature*, Vol. 569, May 2019.
- Gentner, Dedre, and Keith J. Holyoak, "Reasoning and Learning by Analogy: Introduction," *American Psychologist*, Vol. 52, No. 1, 1997.
- Harrison, Katriona, Angus S. Mackay, Lucas Kambanis, Joshua W. C. Maxwell, and Richard J. Payne, "Synthesis and Applications of Mirror-Image Proteins," *Nature Reviews Chemistry*, Vol. 7, May 2023.
- Helmus, Todd C., Liisa Hiatt, and Alejandro Uriel Becerra-Ornelas, *Do National Security Communication Campaigns Work? Taking a Lesson from the Public Health Sector*, RAND Corporation, RR-A766-1, 2022. As of February 25, 2026: https://www.rand.org/pubs/research_reports/RRA766-1.html
- Hutchison, Clyde A., III, Ray-Yuan Chuang, Vladimir N. Noskov, Nacyra Assad-Garcia, Thomas J. Deerinck, Mark H. Ellisman, John Gill, Krishna Kannan, Bogumil J. Karas, Li Ma, et al., "Design and Synthesis of a Minimal Bacterial Genome," *Science*, Vol. 351, No. 6280, March 2016.
- Institut Pasteur, "Mirror Life: Inside the Discussion," July 22, 2025.
- Kalra, Nidhi, Forrest W. Crawford, Matthew L. Nicotra, Scott Behmer, Josh Dettman, Gerald L. Epstein, Edward Geist, David Glickstein, Graham Griffin, and Krista Langeland, *A U.S. Strategy to Prevent the Creation of Mirror Life*, RAND Corporation, RR-A4335-1, forthcoming.
- Kousky, Carolyn, and Leonard Shabman, *Understanding Flood Risk Decision Making: Implications for Flood Risk Communication Program Design*, Resources for the Future, Discussion Paper 15-01, 2015.
- Kubota, Taylor, "A New Report Warns of Serious Risks from 'Mirror Life,'" *Stanford Report*, December 16, 2024.
- Morgan, M. Granger, Baruch Fischhoff, Ann Bostrom, and Cynthia Atman, *Risk Communication: A Mental Models Approach*, Cambridge University Press, 2001.
- National Academies of Sciences, Engineering, and Medicine, "Mirror Image Biology: Pushing the Envelope in Designing Biological Systems—A Workshop," webpage, undated. As of December 1, 2025: <https://www.nationalacademies.org/projects/DELS-BLS-25-03>
- National Academies of Sciences, Engineering, and Medicine, *Building Communication Capacity to Counter Infectious Disease Threats: Proceedings of a Workshop*, National Academies Press, 2016.
- National Institutes of Health, "Stop the Spread of Superbugs: Help Fight Drug-Resistant Bacteria," *NIH News in Health*, February 2014.

National Science Foundation, “Collaborative Research: Theoretical and Experimental Investigations of Soft Colloidal Materials,” webpage, award 1443228, July 31, 2014a. As of July 21, 2025: https://www.nsf.gov/awardsearch/showAward?AWD_ID=1443228&HistoricalAwards=false

National Science Foundation, “ERASynBio: Establishment of a Fully Synthetic, Mirror-Image Biological System,” webpage, award 1443228, July 31, 2014b. As of July 21, 2025: https://www.nsf.gov/awardsearch/showAward?AWD_ID=1443228&HistoricalAwards=false

National Science Foundation, “Collaborative Research: Booting Up a Mirror Cell,” webpage, award EF-1935372, September 2019. As of July 21, 2025: https://www.nsf.gov/awardsearch/showAward?AWD_ID=1935372&HistoricalAwards=false

NSF—See National Science Foundation.

Paris Conference on Risks from Mirror Life, *Paris Conference on Risks from Mirror Life: Meeting Report*, September 2025.

Parker, Andrew M., “The Need to Invest in Social Science Infrastructure to Address Emerging Crises,” in Aaron B. Frank and Elizabeth M. Bartels, eds., *Adaptive Engagement for Undergoverned Spaces: Concepts, Challenges, and Prospects for New Approaches*, RAND Corporation, RR-A1275-1, 2022. As of April 24, 2026: https://www.rand.org/pubs/research_reports/RRA1275-1.html

Parker, Andrew M., Amanda F. Edelman, Katherine G. Carman, and Melissa L. Finucane, “On the Need for Prospective Disaster Survey Panels,” *Disaster Medicine and Public Health Preparedness*, Vol. 14, No. 3, June 2020.

Peplow, Mark, “How Should ‘Mirror Life’ Research Be Restricted? Debate Heats Up,” *Nature*, September 15, 2025.

Pidgeon, Nick, “Engaging Publics About Environmental and Technology Risks: Frames, Values and Deliberation,” *Journal of Risk Research*, Vol. 24, No. 1, January 2021.

Scheerer, Victoria, Chen-Hsu Yu, Xuan Han, and Jonathan T. Sczepanski, “The Clinical Potential of L-Oligonucleotides: Challenges and Opportunities,” *Chemical Science*, Vol. 15, 2024.

Service, Robert F., “Leading Scientists Warn Against Developing ‘Mirror-Image’ Bacteria,” *Science Insider*, December 12, 2024.

Siegrist, Michael, “Trust and Risk Perception: A Critical Review of the Literature,” *Risk Analysis*, Vol. 41, No. 3, March 2021.

Slovic, Paul, “Perception of Risk,” *Science*, Vol. 236, No. 4799, 1987.

Slovic, Paul, “Risk Perception and Risk Analysis in a Hyperpartisan and Virtuously Violent World,” *Risk Analysis*, Vol. 40, No. S1, 2020.

The Spirit of Asilomar and the Future of Biotechnology Summit, “Spirit of Asilomar Entreaty 2025.4.4: Risks from Mirror Life,” Rice University, February 2025.

Statore, Rachel, Aaron Clark-Ginsberg, Shoshana R. Shelton, Nipher Malika, Kristin J. Leuschner, and Tucker Reese, *Community and Individual Disaster Resilience for Floods: Options for Improving Protective Action Guidance*, Homeland Security Operational Analysis Center operated by the RAND Corporation, RR-A1770-1, 2023. As of December 12, 2025: https://www.rand.org/pubs/research_reports/RRA1770-1.html

UK Department for Environment, Food & Rural Affairs, “Ozone Depletion and Climate Change,” webpage, undated. As of September 29, 2025: <https://uk-air.defra.gov.uk/research/ozone-uv/moreinfo?view=depletion-climate-change>

UK Government Office for Science, “Research and Analysis: Mirror Life,” July 16, 2025. As of December 1, 2025: <https://www.gov.uk/government/publications/mirror-life/mirror-life>

University of Manchester, “Manchester Workshop Advances Technical Understanding of Mirror Organism Precursor Technologies,” October 27, 2025. As of May 11, 2026: <https://www.manchester.ac.uk/about/news/manchester-workshop-advances-technical-understanding-of-mirror-organism-precursor-technologies/>

U.S. Centers for Disease Control and Prevention, “About Prion Diseases,” webpage, undated. As of September 29, 2025: <https://www.cdc.gov/prions/about/index.html>

U.S. Department of Agriculture, National Invasive Species Information Center, “What Are Invasive Species?” webpage, undated. As of September 29, 2025: <https://www.invasivespeciesinfo.gov/what-are-invasive-species>

U.S. Environmental Protection Agency, “Risk Messaging Guidance,” last updated July 10, 2025. As of April 24, 2026: <https://www.epa.gov/risk/risk-messaging-guidance>

U.S. Food and Drug Administration, “Agricultural Biotechnology,” webpage, undated. As of March 9, 2026: <https://www.fda.gov/food/consumers/agricultural-biotechnology>

Vonnegut, Kurt, Jr., *Cat's Cradle*, Holt, Rinehart and Winston, 1963.

Wachinger, Gisela, Ortwin Renn, Chloe Begg, and Christian Kuhlicke, "The Risk Perception Paradox—Implications for Governance and Communication of Natural Hazards," *Risk Analysis*, Vol. 33, No. 6, 2013.

Wallsten, Thomas S., David V. Budescu, Rami Zwick, and Steven M. Kemp, "Preferences and Reasons for Communicating Probabilistic Information in Verbal or Numerical Terms," *Bulletin of the Psychonomic Society*, Vol. 31, No. 2, 1993.

Warren, George W., and Sarah M. Duckett, "Risk Perception and Communication, Plugged-In: Fifty Years of Process," *Risk Analysis*, Vol. 46, No. 3, 2026.

Zentrale Kommission für die Biologische Sicherheit, "Spiegelbakterien," September 22, 2025. As of December 1, 2025:
<https://zkbs-online.de/synthetische-biologie/spiegelbakterien#>

About the Authors

Andrew M. Parker is a senior behavioral and social scientist at RAND and a professor of policy analysis at the RAND School of Public Policy. His research applies core concepts in behavioral decision research to the understanding of decisionmakers' behavior in complex, real-world situations, including in health and emergency preparedness. He holds a Ph.D. in behavioral decision theory.

Forrest W. Crawford is a senior statistician at RAND. He conducts technical and policy research on emerging technology and biosecurity. Crawford holds a Ph.D. in biomathematics.

Gerald L. Epstein is an adjunct senior physical scientist at RAND and has spent more than 40 years inside and outside government researching, analyzing, formulating, and communicating public policy at the intersection of science, technology, and security. He works with the Center for AI, Security, and Technology on the governance of biotechnology and the life sciences. He holds a Ph.D. in physics.

Sella Nevo is a senior information scientist at RAND and the director of the Center for AI, Security, and Technology. He conducts technical and policy research on biosecurity and AI security. Nevo holds a dual B.S. degree in computer science and mathematics.

David Glickstein is a research communications analyst at RAND. He holds a Master of Public Policy with a concentration in science and technology policy.

Matthew L. Nicotra is a senior physical scientist at RAND. His research focuses on biosecurity, emerging biotechnology risks, and the intersection of life sciences with national security. Nicotra holds a Ph.D. in ecology and evolutionary biology.

Henry H. Willis is a senior policy researcher at RAND and the associate director of the Center for AI, Security, and Technology. His recent work analyzed biosecurity risks and biodefense capabilities, emerging and existential risks, climate and natural disaster risks, critical infrastructure resilience, and national preparedness to chemical, biological, radiological, and nuclear attacks. He holds a Ph.D. in engineering and public policy.

About This Paper

Mirror life is an alternate form of life in which every molecule is replaced with its mirror image. Recently, a group of scientists warned that mirror life, if created and released, could have devastating consequences for humans, plants, and animals, potentially leading to human extinction. Mirror life does not exist yet, but research programs have made progress in synthesizing mirrored biomolecules, which is an essential step toward creating mirror life, and scientists believe that a living mirror bacterium could be created within a few decades. As scientists and policymakers consider policy options to address the risks from mirror life, they face challenges in communicating these risks to the public, businesses, researchers, and government agencies that could be tasked with implementing policies to respond to the threat.

In this paper, we provide recommended communication strategies for explaining mirror life, its risks, and the policies that government decisionmakers might choose to employ to guard against those risks. We provide recommended language for describing the science of mirror life; describe the potential strengths and pitfalls of drawing analogies to other kinds of threats in communications; and offer ways of addressing confusion and concerns about mirror life, research into mirror life, and governance proposals. Our recommendations are based on technical findings about the risks of mirror life and research on effective risk communication about threats to public health, safety, and security.

Center on AI, Security, and Technology

RAND Global and Emerging Risks is a division of RAND that delivers rigorous and objective public policy research on the most consequential challenges to civilization and global security. This work was undertaken by the division's Center on AI, Security, and Technology, which aims to examine the opportunities and risks of rapid technological change, focusing on artificial intelligence, security, and biotechnology. For more information, contact cast@rand.org.

Funding

This effort was independently initiated and conducted within the Center on AI, Security, and Technology using income from operations and gifts and grants from philanthropic supporters. A complete list of donors and funders is available at www.rand.org/CAST. RAND clients, donors, and grantors have no influence over research findings or recommendations.

RAND is a research organization that develops solutions to public policy challenges to help make communities throughout the world safer and more secure, healthier and more prosperous. RAND is nonprofit, nonpartisan, and committed to the public interest.

Research Integrity

Our mission to help improve policy and decisionmaking through research and analysis is enabled through our core values of quality and objectivity and our unwavering commitment to the highest level of integrity and ethical behavior. To help ensure our research and analysis are rigorous, objective, and nonpartisan, we subject our research publications to a robust and exacting quality-assurance process; avoid both the appearance and reality of financial and other conflicts of interest through staff training, project screening, and a policy of mandatory disclosure; and pursue transparency in our research engagements through our commitment to the open publication of our research findings and recommendations, disclosure of the source of funding of published research, and policies to ensure intellectual independence. For more information, visit www.rand.org/about/research-integrity.

RAND's publications do not necessarily reflect the opinions of its research clients and sponsors. **RAND®** is a registered trademark.

Limited Print and Electronic Distribution Rights

This publication and trademark(s) contained herein are protected by law. This representation of RAND intellectual property is provided for noncommercial use only. Unauthorized posting of this publication online is prohibited; linking directly to its webpage on rand.org is encouraged. Permission is required from RAND to reproduce, or reuse in another form, any of its research products for commercial purposes. For information on reprint and reuse permissions, please visit www.rand.org/about/publishing/permissions.

For more information on this publication, visit www.rand.org/t/PEA4619-1.

© 2026 RAND Corporation



www.rand.org